

SVKM's NMIMS
Mukesh Patel School of Technology Management & Engineering

Program: B Tech, MBA Tech and B Tech Integrated Data Science				Semester: IV/ VIII	
Course: Stochastic processes and Applications				Code: 702BS0C033	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks- 100)
1	0	2	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Statistics and Probability, Calculus					
Course Objective Stochastic Processes and Applications refers to skills, practices and techniques used in stochastic systems found in several real life applications. This discipline consists of an understanding of stochastic systems and their applications in wide array of business situations					
Course Outcomes After completion of the course, students will be able to - 1. Define various aspects of stochastic processes 2. Apply their potential applications in different business domains like manufacturing retail, marketing, operations 3. Analyze various processes in stochastic domain 4. Build transition, distribution and classification problems					
Detailed Syllabus					
Unit	Description				Duration
1.	Probability Theory Basics Axiomatic construction of probability spaces, random variables and vectors, probability distributions, functions of random variables, mathematical expectations, transforms and generating functions, modes of convergence of sequences of random variables, laws of large numbers, central limit theorem, Software for Probability Theory				04
2.	Introduction to Stochastic Process Definition and examples of stochastic processes, classification of random processes according to state space and parameter space, types of stochastic processes, elementary problems, Software for Stochastic Process				04
3.	Stationary Process Weakly stationary and strongly stationary processes, moving average and auto regressive processes, Software for Stationary Process				02



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4.	Discrete-time Markov Chains Definition and examples of markov chains, transition probability matrix, Chapman-Kolmogorov equations; calculation of n-step transition probabilities, limiting probabilities, classification of states, ergodicity, stationary distribution, transient markov chains, random walk and gambler's ruin problem, applications, Software for Discrete-time Markov Chains	03
5.	Continuous-time Markov Chains Kolmogorov-Feller differential equations, infinitesimal generator, Poisson process, birth-death process, stochastic Petri net, Applications, Software for Continuous-time Markov Chains	02
	Total	15
Text Book 1. J. Medhi, <i>Stochastic Processes</i> , 5 th Edition, New Age International, 2017.		
Reference Books 1. Liliana Blanco Castaneda, Viswanathan Arunachalam and S. Dharmaraja, <i>Introduction to Probability and Stochastic Processes with Applications</i> , Wiley, 2019. 2. Suresh Chandra, S. Dharmaraja, Aparna Mehra, R. Khemchandani, <i>Financial Mathematics: An Introduction</i> , Narosa Publication House, 2020. 3. Kishor S. Trivedi, <i>Probability, Statistics with Reliability, Queueing and Computer Science Applications</i> , 5th Edition, Wiley, 2020 .		
Tutorial Work 8 to 10 Tutorial exercises based on the syllabus.		



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Program: B Tech, MBA Tech and B Tech Integrated Data Science				Semester : IV/ VIII	
Course: Statistical Structures in Data and Inference				Code: 702DB0C006	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks- 100)
3	2	0	4	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Managing Uncertainty (SEM - III)					
Course Objective This course aims at giving the foundation knowledge Statistical Inference and regression analysis. In particular, it gives details of theory of Estimation and testing of hypothesis. Both theoretical aspect will be discussed and practical problems will be dealt with in great detail. In this course the students also learn different models for categorical data such as Generalized Linear, logit, logistic and log linear models					
Course Outcomes After completion of the course, students will be able to - 1. Define parametric and non-parametric models 2. Interpret the notion of a parametric models, point and interval estimation of the parameters of those models 3. Apply Linear, Logistic and Non-linear regression models and its implementation in real life situation					
Detailed Syllabus					
Unit	Description				Duration
1.	Statistical Inference • Confidence intervals • Likelihood function and maximum likelihood • Computing the MLE • Computing the MLE: examples • Plotting the likelihood • Inference example: frequent • Inference example: Bayesian • Continuous version of Bayes' theorem • Posterior intervals				08



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2.	Hypothesis Testing - Basics of Estimating the populations mean and difference - Estimating the proportion and difference - Large sample test for population mean, difference - Large sample test for proportion, difference. - Sampling distribution of variance, estimation. - Inferences from small sample: Student's t distribution; Small sample t test for following – A population mean, A difference between two means, Confidence interval. - Rejection and Non-rejection region - Type I and Type II errors - testing hypothesis about a population mean using the Z- statistic - using p-values to test Hypothesis	10
3.	ANOVA/MANOVA - Chi-Square as a test of independent - Chi-square as a Test of goodness of fit: Testing the Appropriateness of a Distribution, Analysis of Variance, Multivariate analysis of variance	03
4.	Regression Model and its Inference - Least squares and linear regression: Introduction; Notation; Ordinary least squares; Regression to the mean; Linear regression; Residuals; Regression inference - Multivariable regression: Multivariate regression; Multivariate examples; Adjustment; Residual variation and diagnostics; Multiple variables, Interaction Terms, Non-linear Transformations of the Predictors, Qualitative Predictors - Multiple Regression Analysis: The Problem of Estimation and the Problem of Inference - Dummy Variable Regression Models - Multi-collinearity, Heteroscedasticity, Autocorrelation - Econometric Modelling: Model Specification and Diagnostic Testing - Correlation and Covariance Analysis - Canonical Analysis, Canonical Roots/variates	12
5.	Extension of regression analysis - Ridge Regression, The Lasso - Nonlinear Regression Models: - Approaches to Estimating Nonlinear Regression models	04



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6.	Generalized linear models - Logistic Regression - Binary outcomes - Count outcomes - Multiple Logistic Regression	04
7.	Nonparametric Test - Sign Test - Rank Sum Test - Kruskal-Wallis and Friedman tests	04
Total		45
Text Books - Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, <i>An Introduction to Statistical learning with application in R</i>, Springer, 2013. - Damodar Gujarati, <i>Basic Econometrics</i>, 5th edition, McGraw Hill Education, 2017.		
Reference Books - Agresti , <i>An Introduction to Categorical Data Analysis</i>, 3rd Edition, Wiley, 2018. - Trevor Hastie, Robert Tibshirani, and Jerome Friedman, <i>The Element of Statistical Learning, Data mining, Inference and Prediction</i>, Springer, 2013.		
Laboratory Work 8 to 10 programming exercises (and a practicum) based on the syllabus		



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Program: B Tech, MBA Tech and B Tech Integrated Data Science, B Tech [AI and ML, AI and DS, CSE (DS)]				Semester : IV / V/ VII/ VIII	
Course: Data Handling and Visualization				Code: 702DB0C001	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks- 100)
2	0	0	2	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Nil					
Course Objective To learn data handling using appropriate tools. To visualize the data using BI tools by understanding various factors involved in developing the charts. To train on using various visual charts for unbiased and effective conveying of information					
Course Outcomes After completion of the course, students will be able to - 1. Choose data precisely and handle it as per the requisite 2. Show the right chart for representing the data 3. Construct and prepare interactive dashboards					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to Data Literacy How to read and interpret data and charts.				03
2.	Understanding color schemes and other rules of creating charts				03
3.	Introducing BI tools - Identify different types of source data, Identify how data fields differ by type and use, Describe the importance of knowing the granularity of data, Demonstrate an understanding of Analytical terminology, Describe key areas of the BI tool workspace, Build common views, including a bar chart and crosstab, Bring views together in an interactive dashboard.				04
4.	Basic Working with BI tools - Joining tables and using calculations, Using multiple data source, Filtering on Data source, Using sets.				05



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5.	Advanced Working with BI tools – Applying context filter, Splits, Table calculations, Secondary table calculations, Level of Details.	04
6.	Analytical Working with BI tools – Using parameters, Advanced mapping, Time based data analysis, Seasonality.	04
7.	Creating various maps to show information	04
8.	Merging BI tools with programming language for enhanced data analytics	03
Total		30

Text Books

1. Cole Nussbaumer Knaflic, *Storytelling with Data: A Data Visualization Guide for Business Professionals*, 1st Edition, Wiley, 2015.
2. Andy Cotgreave, Jeffrey Shaffer and Steve Wexler, *The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios*, 2nd Edition, Wiley, 2017.

Reference Books

1. Stephen Few, *Information Dashboard Design: Displaying Data for At-a-glance Monitoring*, 2nd Edition, Analytics Press, 2013.
2. Scott Berinato, *Good Charts: The HBR Guide to Making Smarter, More Persuasive Data Visualizations*, 1st Edition, Harvard Business School Publishing, 2016.



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Program: B Tech, MBA Tech and B Tech Integrated Data Science, B Tech CSE (DS)				Semester: IV/ VIII	
Course: Introduction to Data, Signal and Image Analysis				Code : 702DB0C007	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks- 100)
3	2	0	4	Marks Scaled to 50	Marks Scaled to 50
Prerequisite: Nil					
Course Objective The interaction of data science and technology with the world is via signal processing: detecting, transcoding, understanding and generating time-dependent and space-dependent signals in the broadest sense. Data is rich and abundant. However, every data wrangler has come across most of the time this data, which is very noisy in nature. Processing noisy signals and filtering unwanted part from an image has been a huge concern over the last few decades. Signal and image processing techniques developed today are very robust and effective models. The objective is to port some processing techniques from the audio and signal field and use them to process data. To give basic mathematical understanding of transformation for image and signal processing and sampling To introduce elementary concepts of LTI system, convolution.					
Course Outcomes After completion of the course, students will be able to - 1. Classify discrete and continuous data signals and learn the engineering mathematical aspect of it and perform operations on it 2. Model finite impulse and infinite impulse response filters 3. Simplify the concepts of Convolution and Sampling to solve and support problems 4. List the fundamental concepts in Image representation and state basic filtering techniques to facilitate noise removal					
Detailed Syllabus					
Unit	Description				Duration
1.	Data Communication Fundamentals, Data and signals, A simplified model of a data communication system, Defining Data and its types, Electrical representation of data, its characteristics, domain and spectrum concept, Distinguish between data and signal, basic parameters that are required for characterizing the signal, introduction to standard cts, mathematical operations on cts				08
2.	LTI convolution (Discrete), Convolution integral (continuous), single and double side representation of signal, circular (periodic) convolution (for				08



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	discrete), Z-transform; finding z-tx. of signal, problems on ROC, using ROC problem determining whether system is stable, causal or anti-causal, sampling problem without proof to determine min/ideal/nyquist sampling rate when an analog signal is given, finding transfer function, impulse response, step response from given difference equation	
3.	Fourier Series (for periodic cts); Trigonometric; exponential, Fourier Transform of signal(non-periodic continuous); properties with proof, Laplace transform; its properties, problems on region of convergence, from given equation finding zero input; state and total response, problems of initial and final values	12
4.	Digital iir filter design; impulse invariant; bilinear transformation, Butterworth filter design, fir filter design using window function, fir vs. iir	08
5.	Image Representation; What is a digital image; its layout; color, resolution and quantisation, bit plane slicing, image formats, image data types, image compression, color spaces	04
6.	What is Pixel; Histogram equalisation and matching, why perform enhancement; filtering for noise removal; mean; median; rank; Gaussian filtering	05
	Total	45

Text Books

1. Nagoor Kani, *Signals and Systems*, Mc Graw Hill Education India pvt. Ltd., 20th Edition, 2018.
2. Anil Jain, *Fundamentals of Digital Images Processing*, 1st Edition, Pearson, 2015.

Reference Book

1. S. Salivahanan, *Digital Signal Processing*, 4th Edition, McGraw Hill, 2019.

Laboratory Work

8 to 10 programming exercises (and a practicum) based on the syllabus



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Program: B Tech (Artificial Intelligence, Data Science, Computer Engineering, Information Technology, CSE (Cyber), AI and ML, AI and DS, CSBS, CSE (DS), Computer Science) MBA Tech (All Programs) B Tech Integrated (Data Science, Computer Engineering, Information Technology)				Semester: III/IV/ VII/ VIII	
Course: Database Management Systems				Code: 702AI0C001	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Nil					
Course Objective The objective of the course is to provide a comprehensive introduction to the fundamental concepts for design and development of database systems, with an emphasis on how to organize, maintain and retrieve - efficiently, and effectively - information from a database management system					
Course Outcomes After completion of the course, students will be able to - 1. Describe core concepts of database and model a database management system through ER modelling 2. Apply knowledge of relational algebra and structured query language to retrieve and manage data from relational database 3. Demonstrate the use of normalization for database design 4. Demonstrate the concept of transactions and use modern database techniques such as NoSQL					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction Database System Applications, Purpose of Database Systems, View of Data, Database Languages, Data Models, Database Users and Administrator				03
2.	Database Design and the E-R Model Overview of the Design Process, The Entity-Relationship Model, Constraints, Entity Relationship Diagrams, Reduction to Relational Schemas, Schema Diagrams, Entity-Relationship Design Issues, Extended ER features				05
3.	Introduction to the Relational Model Structure of Relational Databases, Database Schema, Keys, Relational Algebra, Basic operators of Relational Algebra,				03
4.	Structured Query Language Overview of the SQL Query Language, SQL Data Definition, SQL Constraints, Basic Structure of SQL Queries, Additional Basic Operations, DML operations, Set operations, Aggregate Functions, Nested Sub-queries, Joins, views				06



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5.	Relational Database Design Features of Good Relational Designs, Problems with bad design, Decomposition using concept of functional dependencies, Armstrong's axioms, Closure of functional dependency, Closure of attribute, Introduction to process of Normalization and de-normalization, Normal Forms- 1NF, 2NF, 3NF, BCNF	05
6.	Transactions What is Transactions? Properties of transaction, Transaction states, Issues with concurrent executions, Schedules, Serializability- Conflict and View	04
7.	Introduction to NoSQL Overview of NoSQL, characteristics of NoSQL, Storage types of NoSQL, Implementing NoSQL in MongoDB - Managing Databases and Collections from the MongoDB shell, Finding Documents in MongoDB collection from the MongoDB shell.	04
	Total	30
Text Books		
1. Henney Korth and Abraham Silberschatz, <i>Database System Concepts</i> , 7 th Edition, McGraw Hill, 2019. 2. Elmasri and Navathe, <i>Fundamentals of Database Design</i> , 7 th Edition, Addison Wesley, 2019. 3. A Phaltankar, J. Ahsan, M. Harrison, L. Nevdov, <i>MongoDB Fundamental</i> , Packt Publishing, 2020.		
Reference Books		
1. Bob Bryla, Kevin Loney <i>Oracle Database 12C The Complete Reference</i> , 1 st Edition, Tata McGraw Hill, 2017. 2. Marko Aleksendric, Arek Borucki, <i>Mastering MongoDB 7.0</i> , Packt Publishing, 2024.		
Laboratory Work		
8 to 10 experiments (and a practicum where applicable) based on the syllabus		



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Program: B Tech (Artificial Intelligence, Data Science, Computer Engineering, Information Technology, AI and ML, AI and DS, CSE (Cyber), CSE (DS)) MBA Tech (All Programs) B Tech Integrated (Data Science, Computer Engineering)				Semester: IV/V/VI/ V-VII/ VIII/ IX	
Course: Mobile Application Development				Code: 702AI0E002	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 100)	Term End Examinations (TEE)
2	2	0	3	Marks Scaled to 100	-
Prerequisite: Knowledge of Programming					
Course Objective The objective of this course is to gain insights into the Android and IOS Operating systems and to understand the components and layouts of these applications. It will also help the students to implement database connectivity with real-time databases and further develop an Android/IOS based application.					
Course Outcomes After completion of the course, students will be able to - 1. Design user interfaces using Android Studio and Flutter 2. Implement file handling using text and images 3. Implement database connectivity and location tracking 4. Develop a full-fledged Android/IOS application					
Detailed Syllabus					
Unit	Description				Duration
1.	Configuration of Development Platform Starting an Android Application project/IOS Application Project: Installing the Application Development Kit (Android Studio / IOS)				02
2.	Understanding the different Components for Application Design Screen Layout, Simple Controls, Creating and Configuring an Android Emulator, Communicating with the Emulator. Controls and the User Interface: Check Boxes, Radio Buttons, Spinner, Date Picker, Touch Listener, Graphics. Multiscreen Applications: Stretching the Screen, Pop-up Dialog Boxes and Toasts, Menus.				10
3.	Inputting Images and File Handling Displaying Images, Using Images stored on the Android Device, File handling using .txt and .csv files				04
4.	Location Tracking Location Tracking using Google maps				02
5.	Introduction to Flutter Understanding the configuration and UI development using Flutter				04
6.	Processing using Databases Database connectivity using SQLite 3 and Firebase				05



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7.	Application Publishing Client-Server Applications and Publishing your application	03
	Total	30
Text Books		
1. John Horton, <i>Android Application Development for Java Programmers</i>, 3rd Edition, Packt Publishing, 2021. 2. Barry Burd, <i>Flutter for Dummies</i>, 1st Edition, 2020.		
Reference Books		
1. Barry Burd, <i>Android Application Development All in one for Dummies</i>, 3rd Edition, July 2020. 2. Rick Boyer, <i>Android 9 Development Cookbook</i>, 3rd Edition, Packt Publishing, 2018. 3. Alessandro Biessek, <i>Flutter for Beginners</i>, 1st Edition, Packt Publishing, 2019.		
Laboratory Work		
8 to 10 experiments (and a practicum where applicable) based on the syllabus		



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